

# Creating EQuilibrium in Canada

**T**welve Canadian home builder teams were recently selected as winners of Canada Mortgage and Housing Corporation (CMHC)'s EQuilibrium sustainable housing competition. The competition was organized by CMHC to support the construction of sustainable, healthy houses that are also affordable and energy and resource efficient. EQuilibrium housing is designed to lower homeowners' energy bills by reducing energy consumption and delivering electricity back to the grid. The homes will also promote water conservation, healthy indoor environments, durability, and reduced pollutant emissions.

Each winning team will receive \$50,000 from CMHC to offset eligible costs, including those relating to documenting the projects, performance testing, and demonstrating the homes publicly. In addition, CMHC will work with the winning teams to provide technical and promotional support, and will monitor and report on the performance of the houses.

The winning projects were chosen by independent housing experts from a total of 72 home builder teams that submitted applications in July 2006 through a two-stage competitive process. The demonstration homes will be open to Canadians to view by 2008, after which CMHC will explore opportunities to advance EQuilibrium housing principles more broadly across the housing industry.

"These houses are designed to produce as much energy as they consume on an annual basis. They will be a blueprint for the next generation of housing in Canada," says the Honourable Monte Solberg, Canada's minister of human resources and social development, and the minister responsible for CMHC.

## The Winning Proposals

The Abondance Montréal will be built in Verdun, Québec, by Ecocité Developments. The triplex will draw energy from several sources, including a

GeoExchange heat pump system, PV panels, and solar-thermal vacuum tubes (that is, evacuated tube collectors). It features toilets that run on captured rainwater and a total of 84 solar panels.

The Alouette Homes EQuilibrium Initiative home will be built in Eastman, Québec. Its design couples readily available renewable energy technologies with energy-efficient construction techniques. The design focuses on using factory preengineered modular sections to

reduce environmental impact at a rural building site due to driving and other costs. The home will be connected to the electrical grid using a net metering system, allowing the eventual owner to "sell" excess electricity generated by the home's PV system to the grid.

Team Montréal Zéro, in Hudson, Québec, will create the EQuilibrium #1, a single-family, detached house to be located in Hudson (see Figure 1). The proposed house will have a well-insulated building envelope and will rely heavily on passive-heating and -cooling techniques to achieve its EQuilibrium target. In order to ensure air quality, careful consideration will be given to design and construction techniques that protect and enhance the indoor environment. A large portion of the site will remain undisturbed and will act as a natural wildlife habitat.

Ampas Architect in Toronto, Ontario, has proposed the Sustainable Urbanism Initiative, which is comprised of three town houses located in Toronto's downtown Annex area. The town houses will incorporate energy-efficient features such as spectrally selective window glazing and will incorporate ground



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source heat pumps powered by electricity generated through PV cells covering the roof of each unit. The town houses' urban location will reduce dependence on the automobile and encourage efficient use of existing infrastructure.

Minto, from Ottawa, Ontario, will construct the Minto Manotick House in the south-end of Ottawa. The house's envelope includes a high level of insulation, double-wall construction, and triple-pane windows. The design incorporates an all-off switch that turns off all lights, computers, cable boxes, amplifiers, and other miscellaneous electronics.

Lorraine Gauthier's Now House project will take an existing post-World War II house in Toronto and retrofit it to meet EQuilibrium's goals. Insulation upgrades, new windows, Energy Star appliances, wastewater heat recovery, and solar panels will be used to reduce the house's energy usage footprint and greenhouse gas emissions. The larger goal of this plan is to demonstrate how homeowners and their local contractors can improve the energy efficiency of older housing with a few simple but innovative modifications.

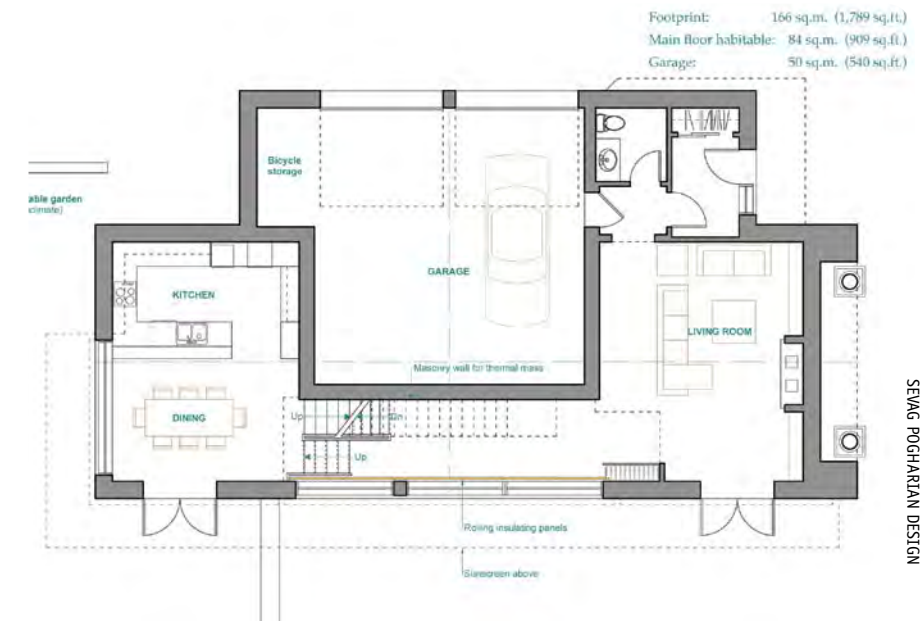
Avalon Master Builder's Discovery III, a grid-tied solar home to be built in

Red Deer, Alberta, is designed to produce as much energy as it uses annually. It will also feature a graywater recycling system to reduce the home's reliance on the municipal water system. The home will also use solar systems to preheat air and water.

Canadian Housing Energy Sustainable Solutions (CHESS), in Red Deer, Alberta, is building the EQUilibrium Concept Home, which is customized for Central Alberta's moderate climate—specifically, for that of Red Deer. It pays close attention to the issue of resource efficiency. Not only will 65% of the construction waste generated by the home be recycled, but many of the materials used to build it can be reused when the home is eventually demolished. The home is also intended to evolve with the owner's needs. For example, an optional second floor can be developed to accommodate a growing family. Later in life, the main floor can easily be adapted to be barrier-free.

Echo-Logic describes its Echo Haven project as "the next step" in the demonstration of energy-efficient, healthy, low-impact housing on a community scale. The project covers the construction of 25 community homes in Calgary. The project's other features include a greenhouse, a community building with guest accommodation, and a work-at-home office. The design of the homes will demonstrate healthy and durable materials, rainwater harvesting, composting or low-flush toilets, and site-sensitive orientation to maximize both solar exposure and integration with surrounding nature.

Habitat Studio & Workshop in Edmonton, Alberta, has proposed the Riverdale project with the goal of proving that it is possible to build houses that foster a high quality of life while also reducing greenhouse gas production and environmental impact. Riverdale involves the construction of an EQUilibrium-compliant energy duplex, which will produce as much energy as it consumes by combining rigorous energy conservation, energy efficiency measures and the use of renewable energy sources. The materials used to build the home will include regionally pro-



**Figure 1.** The EQUilibrium #1 features thermal mass masonry walls and abundant outdoor space.

duced lumber and recycled newspaper. Riverdale will be located in downtown Edmonton.

The Nexus Solar Corporation and Battleford's Tribal Council project, Yellowhead Innovation Park, Incorporated (YIPI!), will be located in North Battleford, Saskatchewan. The project will feature solar power and the use of low-toxicity construction materials. Nexus has also incorporated innovative refrigeration and clothes-drying technologies into the YIPI! project, as well as factory-built construction that will ensure consistently high quality, affordability, and easy online ordering.

The Winnipeg Housing Rehabilitation Corporation in Winnipeg, Manitoba, is constructing Urban Ecology, a semidetached, environmentally friendly development in an inner-city Winnipeg neighborhood. To adapt to Winnipeg's climate, the units will feature flat and pitched roofs to keep the PV system snow-free in winter. High-efficiency appliances, low-flow fixtures, and sustainable building materials are also integral to the design.

### Canada's Housing Goals

CMHC is leading the EQUilibrium housing initiative. Key stakeholders include other federal departments,

members of Canada's housing and renewable energy industry, home builders and developers, utilities, manufacturers, architects, and other housing experts. Through the EQUilibrium initiative, CMHC's goal is to mobilize partners to advance new sustainable-housing solutions for healthy communities and a clean-energy future.

EQUilibrium housing combines energy-efficient design with renewable energy systems to minimize energy consumption and reduce environmental impact. Although EQUilibrium housing is a recent initiative, its foundations were laid more than 60 years ago, when CMHC was created. CMHC's record of innovative housing initiatives started in the post-World War II period, and includes housing design, urban renewal and neighborhood improvement, housing rehabilitation and renovation, and sewage treatment assistance.

CMHC has been Canada's national housing agency for more than 60 years. CMHC is committed to helping Canadians access a wide choice of high-quality affordable homes, and to making vibrant, healthy communities a reality across the country.

—Elka Karl

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